Yung-Chin (Jim) Chen

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Research Interests

In-Memory Computing (IMC), ML Accelerators, Computer Architecture, LLM Quantization

Education

Princeton University | ECE Ph.D. Candidate *(Princeton, NJ, USA)* Sep 2024 – Jun 2029 (Expected)

- Advisor: Naveen Verma
- *Coursework: ML Systems · VLSI Design · Domain-Specific Architectures · Reinforcement Learning*

National Taiwan University (NTU) | B.S. in EE *(Taipei, Taiwan)* Sep 2019 – Jan 2024

- Phi Tau Phi Honorary Member (for top 1% of college graduates)
- *Coursework: IC Design & Tapeout · Computer Architecture · Machine Learning · VLSI Signal Processing · FPGA*

Keio University | Exchange Student *(Tokyo, Japan)* Oct 2022 – Jul 2023

Research Experience

Verma Lab, Princeton University · Advisor: Naveen Verma Jan 2025 – Present

- **Quantization Algorithms for Analog IMC** — Developed hardware-aware algorithms for operand clipping [1] and mixed-precision IMC computation [2] to improve LLM inference accuracy.
- **LLM Quantization & Activation Spikes** — Uncovered the mechanism of massive activation spikes in LLMs and leveraged their structural role to enable spike-free post-training quantization [3].

Computing and Sensing Group, Keio University · Advisor: Kentaro Yoshioka Sep 2022 – Feb 2024

- **IMC Circuits & Macros** — Developed saliency-aware IMC circuit designs [6, 7, 12] and explored techniques for efficient/accurate analog IMC circuits [10, 11]; helped incorporate bit-parallel analog input into an analog IMC macro [ISSCC'24 / JSSC'25, acknowledged contribution (non-author)].
- **IMC Architectures & Modeling** — Developed a circuit-aware IMC simulator [5], sparsity-centric IMC architecture [8], and algorithm–architecture co-design for emerging neural networks [9].

Energy-Efficient Circuits and Systems Lab, NTU · Advisor: Tsung-Te Liu Sep 2021 – Feb 2024

- **IMC Accelerator Chip** — Developed and taped out a 28nm end-to-end SRAM IMC processor [4].

Work Experience

EnCharge AI Inc. · Supervisor: Chia-Yu Chen *(Santa Clara, CA)* Aug 2026 – Dec 2026

- **Research Intern** — Developing model-quantization methods for IMC architectures

Publications

[1] **Yung-Chin Chen**, Chia-Yu Chen, Naveen Verma, "IMC-CLINIC: Coupled Loss-Informed Newton Iterations for Clipping in Analog In-Memory Computing", *Under review at ICLR 2027*

[2] **Yung-Chin Chen**, Ourania Stouraiti, Naveen Verma, "SMART-IMC: Sensitivity-Aware Mixed ADC Resolution for Transformer Inference with Analog In-Memory Computing", *Submitted to MLSys 2027*

[3] **Yung-Chin Chen**, Chung Peng Lee, Ze-Wei Liou, Naveen Verma, "Massive Spikes in LLMs are Bias Vectors: Mechanistic Uncovering and Spike-Free Quantization", *in Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026*

[4] Tsung-Yen Wu*, Yi-Shuan Pan*, **Yung-Chin Chen***, Wei-Chun Tseng*, ..., Tsung-Te Liu, "N2N-BP^{2}CIM: A 28nm End-to-End Bit-Parallel, Bit-Parallel Computing-In-Memory DNN Processor with High-bandwidth 14T-SRAM Bitcell", *in IEEE Transactions on Circuits and Systems II: Express Briefs (TCAS-II), 2026* **(* Co-first author)**

[5] Wenlun Zhang, Shimpei Ando, **Yung-Chin Chen**, Kentaro Yoshioka, "ASiM: Modeling and Analyzing Inference Accuracy of SRAM-Based Analog CiM Circuits", *in IEEE Transactions on Very Large Scale Integration Systems (TVLSI), 2025*

[6] Shimpei Ando, Satomi Miyagi, Wenlun Zhang, **Yung-Chin Chen**, Kentaro Yoshioka, "A 4541 TOPS/W Saliency-Aware Analog Computing In-Memory Macro with Charge-Domain Saliency Detector", *in IEEE European Solid-State Electronics Research Conference (ESSERC), 2025*

[7] **Yung-Chin Chen**, Shimpei Ando, Daichi Fujiki, Shinya Takamaeda-Yamazaki, Kentaro Yoshioka, "OSA-HCIM: On-The-Fly Saliency-Aware Hybrid SRAM CIM with Dynamic Precision Configuration", *in Asia and South Pacific Design Automation Conference (ASP-DAC), 2024*

[8] Wenlun Zhang, Shimpei Ando, **Yung-Chin Chen**, Satomi Miyagi, Shinya Takamaeda-Yamazaki, Kentaro Yoshioka, "PaCiM: A Sparsity-Centric Hybrid Compute-in-Memory Architecture via Probabilistic Approximation", *in International Conference on Computer-Aided Design (ICCAD), 2024*

[9] **Yung-Chin Chen**, Shimpei Ando, Daichi Fujiki, Shinya Takamaeda-Yamazaki, Kentaro Yoshioka, "HALO-CAT: A Hidden Network Processor with Activation-Localized CIM Architecture and Layer-Penetrative Tiling", *on arXiv*

[10] Kentaro Yoshioka, Shimpei Ando, Satomi Miyagi, **Yung-Chin Chen**, Wenlun Zhang, "Towards Efficient and Precise Analog Compute-in-Memory Circuits", *in International Conference on Solid State Devices and Materials (SSDM), 2024 (Invited)*

[11] Kentaro Yoshioka, Shimpei Ando, Satomi Miyagi, **Yung-Chin Chen**, Wenlun Zhang, "A Review of SRAM-based Compute-in-Memory Circuits", *in Japanese Journal of Applied Physics (JJAP), 2024*

[12] Shimpei Ando, **Yung-Chin Chen**, Satomi Miyagi, Wenlun Zhang, Kentaro Yoshioka, "A Saliency-Aware Analog Computing In-Memory with SAR-Embedded Detection Achieving 18.5% Power Reduction", *in Japanese Journal of Applied Physics (JJAP), 2024*

Honors and Awards

Irving T. Ho Memorial Scholarship - EE dept. at NTU	Dec 2022
The Memorial Scholarship Foundation to Lin Hsiung Chen (acc. rate: 2%)	Nov 2022
Research Grant - National Science and Technology Council (NSTC), Taiwan	Jul 2022 – Jan 2023
Research Grant - Taiwan Semiconductor Manufacturing Co., Ltd (TSMC)	Feb 2022 – Jun 2022
Principal Award - Awarded to top 2% of NTUEE students for academic excellence	Dec 2022

Technical Skills

Programming: Python, C++, SystemVerilog, Verilog, MATLAB
ML Libraries: PyTorch, TensorFlow, NumPy
EDA & Hardware: Design Compiler, Innovus, Virtuoso, HSPICE, NC-Verilog, FPGA
Systems & Tools: Linux, Git, Slurm, LaTeX

Teaching Experience

Teaching Assistant, Design of VLSI Systems — Prof. Naveen Verma	Sep 2025 – Dec 2025
Teaching Assistant, Computer Architecture — Prof. Tsung-Te Liu	Sep 2023 – Jan 2024
Teaching Assistant, EECS Lab Undergraduate Research — Prof. Tsung-Te Liu	Sep 2023 – Jan 2024
Teaching Assistant, Signals and Systems — Prof. Lin-Shan Lee	Feb 2022 – Jun 2022